Pythagorean Theorem: Mini-Lessons

5.1 Legs, hypotenuse and $a^2 + b^2 = c^2$ NC.8.G.6

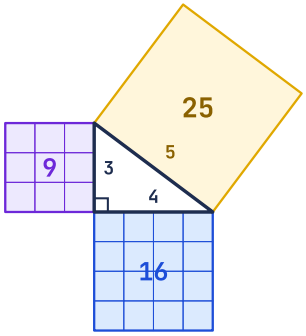
In a **right triangle**, the two shorter sides are the **legs** (a and b). The longest side, across from the right angle, is the **hypotenuse** (c). The theorem: **$a^2 + b^2 = c^2$**.

Right triangle
a triangle with a 90° angle (a square corner)

Leg
one of the two sides that make the square corner

Hypotenuse
the longest side, across from the square corner

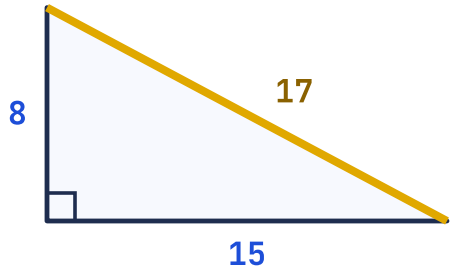
Watch out: The hypotenuse is **always** across from the square corner. Find it first.



$9 + 16 = 25$

$a^2 + b^2 = c^2$

Walk through: Which side is the hypotenuse?

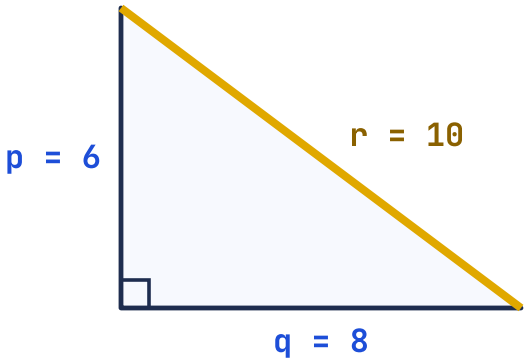


1. Where is the right angle (square corner)?
   Between the sides 8 and 15 → legs: 8 and 15
2. The hypotenuse is across from the right angle. Which side?
   17 → hypotenuse c = 17
3. Check with the theorem: $8^2 + 15^2 = ?$
   $64 + 225 = 289 = 17^2$ → $64 + 225 = 289 = 17^2$ ✓

Check your understanding

1 Score 3 NC.8.G.6

Which side is the **hypotenuse**?



(A) r = 10 (B) p = 6

(C) q = 8

2 Score 3 NC.8.G.6

A right triangle has legs a and b and hypotenuse c. Which equation is true?

(A) $a^2 \cdot b^2 = c^2$ (B) $a^2 + c^2 = b^2$

(C) $a^2 + b^2 = c^2$ (D) $a + b = c$

5.2 Find the hypotenuse NC.8.G.7

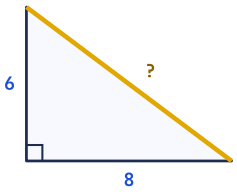
When the **hypotenuse is missing**: square each leg, **add**, then take the **square root**.

Square root (√)

undoes squaring: √100 = 10

Round to the nearest tenth

one digit after the decimal point: √13 ≈ 3.6



$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

$6^2 + 8^2 = c^2$

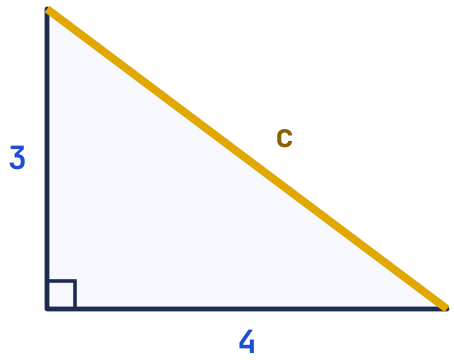
$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

Watch out: Do not add the legs (6 + 8 = 14). Square them first.

Walk through: `Legs 3 and 4. Find the hypotenuse.`

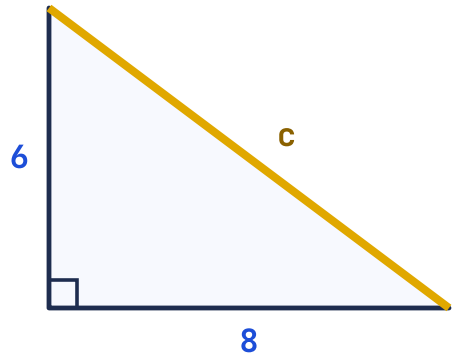


1. Write the theorem with the legs in place.
   $3^2 + 4^2 = c^2$ → $3^2 + 4^2 = c^2$
2. Square and add.
   $9 + 16 = 25$ → $c^2 = 25$
3. $c^2 = 25$. Find c.
   $c = \sqrt{25} = 5$ → $c = 5$

Check your understanding

1 Score 3 NC.8.G.7

The legs of a right triangle are **6** and **8**. Find the hypotenuse.

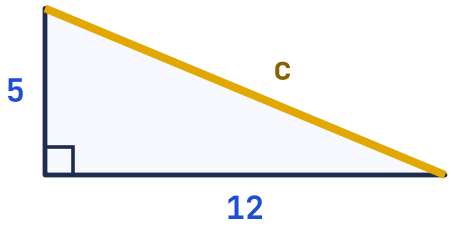


Answer: **10**

1. $6^2 + 8^2 = c^2$
2. $36 + 64 = c^2$
3. $c^2 = 100$
4. $c = \sqrt{100} = 10$

2 Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.



Answer: **13**

1. $5^2 + 12^2 = c^2$
2. $25 + 144 = c^2$
3. $c^2 = 169$
4. $c = \sqrt{169} = 13$

5.3 NC.8.G.7

Find a missing leg

When a **leg is missing**: square the hypotenuse and the leg you know, **subtract**, then take the **square root**. `leg² = c² - (other leg)²`

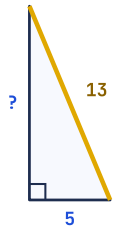
Hypotenuse → add

missing the long side: add the squares

Leg → subtract

missing a short side: subtract the squares

Watch out: If the missing side is a leg, the answer must be **shorter** than the hypotenuse. If it is not, you added by mistake.



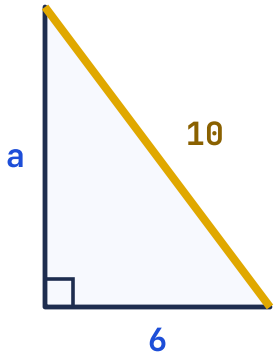
$a^2 + 5^2 = 13^2$

$a^2 + 25 = 169$

$a^2 = 169 - 25 = 144$

$a = \sqrt{144} = 12$

Walk through: `Hypotenuse 10, one leg 6. Find the other leg.`

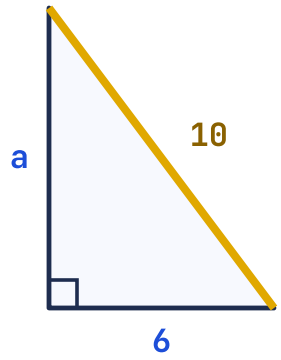


1. Which side is missing?
   A leg: subtract the squares → $a^2 + 6^2 = 10^2$
2. Square and subtract.
   $100 - 36 = 64 \rightarrow a^2 = 64$
3. Find a.
   $a = \sqrt{64} = 8 \rightarrow a = 8$

Check your understanding

1 Score 3 NC.8.G.7

A right triangle has hypotenuse **10** and one leg **6**. Find the other leg.

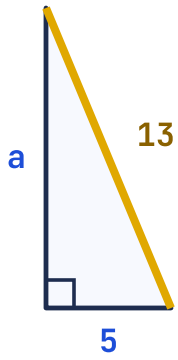


Answer: **8**

1. $a^2 + 6^2 = 10^2$
2. $a^2 = 100 - 36 = 64$
3. $a = \sqrt{64} = 8$

2 Score 3 NC.8.G.7

A right triangle has hypotenuse **13** and one leg **5**. Find the other leg.



Answer: **12**

1. $a^2 + 5^2 = 13^2$
2. $a^2 = 169 - 25 = 144$
3. $a = \sqrt{144} = 12$

5.4 Is it a right triangle? (the converse) NC.8.G.6

To test three side lengths: square the two **shorter** sides and add. If the sum **equals** the longest side squared, it **is** a right triangle. If not, it is not.

Converse
the theorem used backward, to test a triangle

Longest side
the only side that could be the hypotenuse

Watch out: Always use the **longest** side as c.

Sides 6, 8, 10 — $36 + 64 = 100$, $100 = 100$ ✓ — right triangle

Sides 4, 5, 6 — $16 + 25 = 41$, $41 \neq 36$ ✗ — not a right triangle

Walk through: Do 9, 12, 15 make a right triangle?

1. Which side would be the hypotenuse?
   15, the longest side → c = 15
2. Square and add the two shorter sides.
   81 + 144 = 225 → $9^2 + 12^2 = 225$

3 $15^2 = 225$. What do we decide?

Equal, so it IS a right triangle → 225 = 225 → right triangle

Check your understanding

1 Score 3 NC.8.G.6

Can the sides **6, 8, 10** make a right triangle?

- **A** **Yes: 36 + 64 = 100, which is 10^2**
- B No: the numbers are not 3, 4, 5
- C No: 6 + 8 is not 10

1 $6^2 + 8^2 = 36 + 64 = 100$

2 $10^2 = 100$

3 `equal → right triangle`

2 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

- **A** **No: 16 + 25 = 41, but $6^2 = 36$**
- B Yes: 4 + 5 is more than 6
- C Yes: they are in counting order

1 $4^2 + 5^2 = 16 + 25 = 41$

2 $6^2 = 36$

3 `not equal → not a right triangle`

5.5 Distance on the coordinate plane NC.8.G.8

To find the distance between two points: draw a right triangle. The **across** change is one leg, the **up/down** change is the other leg, and the **distance** is the hypotenuse.

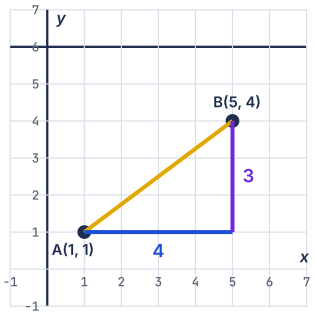
Run

the across change: subtract the x-values

Rise

the up/down change: subtract the y-values

Watch out: Count the length of each leg (always positive). Then use $a^2 + b^2 = c^2$.



`run = 5 - 1 = 4`

`rise = 4 - 1 = 3`

$4^2 + 3^2 = 16 + 9 = 25$

`distance = √25 = 5`

Walk through: `Find the distance from (2, 1) to (8, 9).`

1 Find the run (across change).

8 − 2 = 6 → run = 6

2 Find the rise (up/down change).

9 − 1 = 8 → rise = 8

3 Use $a^2 + b^2 = c^2$.

36 + 64 = 100, so the distance is 10 → distance = √100 = 10

Check your understanding

1 Score 3 NC.8.G.8

Find the distance from **(1, 2)** to **(4, 6)**.

Answer: **5**

1. `run = 4 - 1 = 3, rise = 6 - 2 = 4`
2. $3^2 + 4^2 = 9 + 16 = 25$
3. `distance = √25 = 5`